

FITBUDDY

AI Fitness Plan Generator using Gemini Models

BRAINSTORMING AND IDEATION PHASE

This 10-page document presents the brainstorming and ideation stage of the FitBuddy project. It explains the identified problem, user needs, possible solution ideas, user scenarios, feature concepts, AI and technology direction, concept refinement and the final proposed idea.

Project Area	Generative AI / Fitness / Web Application
Main Objective	Generate personalized workout plans and supporting nutrition or recovery tips
AI Direction	Google Gemini models
Backend Direction	FastAPI
Frontend Direction	HTML + Jinja2
Database Direction	SQLite + SQLAlchemy

Ideation Goal: Convert the identified fitness-planning problem into a practical, user-centered application concept that can generate and update personalized fitness plans.

1. Understanding the Problem

The FitBuddy project begins with the observation that users may have different fitness goals and different preferred workout intensities. The project documentation identifies weight loss, muscle gain and general wellness as examples of fitness objectives.

1.1 User Information Considered

- Name or username
- Age
- Weight
- Fitness goal
- Preferred workout intensity
- Feedback about the existing workout plan

1.2 Problems Identified During Ideation

- A single fixed workout routine may not address different user goals.
- Users may need different levels of workout intensity.
- A plan may need modification when the user provides feedback.
- Workout guidance can be supported by nutrition or recovery advice.
- Stored records can help administrators or coaches review user plans.

1.3 Problem Statement

The ideation problem can be stated as follows: **How can generative AI be used to provide a simple, personalized and adaptable fitness-planning experience through a web application?**

This question guided the brainstorming process toward an application that combines AI-generated content, user input and feedback.

2. User Needs and Expectations

The next part of ideation focused on what a potential user would expect from a fitness-planning application. The requirements were derived from the project scenarios and described functions.

2.1 Need for Personalization

Users should be able to enter their own fitness information and choose a goal and intensity. The generated plan should use these inputs so that the output is relevant to the selected objective.

2.2 Need for Simplicity

The application should provide a straightforward flow: enter information, generate the plan, view the result and optionally provide feedback. The frontend is therefore planned around clear HTML forms and result pages.

2.3 Need for Adaptability

The brainstorming process identified feedback as an important part of the concept. Instead of creating only one fixed plan, the user can request changes such as more cardio or additional rest days.

2.4 Need for Supporting Guidance

The project also identifies nutrition or recovery tips as a complementary feature. This provides a small amount of practical guidance alongside the workout plan.

2.5 Need for Record Keeping

The application concept includes storing user details and generated plans. The documentation also specifies retaining original and updated versions of plans.

3. Brainstorming of Possible Solutions

Several ideas were considered during the brainstorming stage. These ideas were later combined into the FitBuddy concept.

No.	Idea	Purpose
1	Personalized Workout Generator	Generate a structured 7-day workout plan.
2	Nutrition / Recovery Tip	Give concise goal-related supporting advice.
3	Feedback-Based Updating	Revise the workout plan based on user feedback.
4	User Data Storage	Store user information and generated plans.
5	Admin Dashboard	Allow administrators or coaches to view records.
6	Responsive Web Interface	Make the system easy to use on different screens.

3.1 Brainstorming Questions

- What information should the user provide?
- What type of workout output should AI generate?
- How can the plan be structured so that it is easy to follow?
- How can user feedback change an existing plan?
- What supporting nutrition or recovery information is useful?
- How should original and updated plans be stored?
- How can an administrator view the stored information?

4. User Scenarios and Idea Validation

The project documentation provides four scenarios. These scenarios helped validate whether the brainstormed features formed a complete application flow.

Scenario 1 – Personalized Workout

The user enters details including name, age, weight, fitness goal and preferred intensity. The system generates a personalized 7-day workout plan with a day-by-day structure.

Scenario 2 – Feedback and Revision

The user reviews the existing plan and submits feedback such as requesting more cardio or more rest days. The system uses the original plan and feedback to create an updated plan.

Scenario 3 – Nutrition or Recovery Tip

The system provides a concise suggestion related to the user's selected goal. Examples in the documentation include hydration, protein suggestions and recovery practices.

Scenario 4 – Admin or Coach View

An administrator or coach can view registered users and their workout plans. The concept allows comparison of original and updated plans.

Scenario Validation

Together, the scenarios cover input, AI generation, supporting advice, feedback, updating and storage. This helped refine the project from a simple workout generator into a more complete interactive fitness-planning system.

5. Feature Ideation and Requirement Formation

The brainstormed ideas were translated into major functional requirements. Each requirement represents an expected capability of the proposed FitBuddy application.

Feature	Requirement
User Input	Collect username, user ID, age, weight, goal and intensity.
Workout Generation	Create a structured 7-day plan with warm-up, workout and cooldown/recovery guidance.
Nutrition Tip	Generate a concise nutrition or recovery suggestion.
Feedback	Accept user feedback and use it to revise the existing plan.
Plan Storage	Save original and updated plans.
Admin View	Display users and their plans in a structured view.
Frontend	Provide clear pages for input, results and administrative viewing.

5.1 Expected Workout Output

- Day-by-day organization
- Workout focus such as full body, upper body, lower body, cardio, core or flexibility
- Exercise names
- Sets and repetitions or duration
- Suggested rest intervals
- Warm-up and cooldown/recovery guidance

6. AI and Technology Ideation

The ideation process also considered which technologies could support the proposed features. The project documentation describes experimentation with several models before selecting Gemini.

6.1 Models Considered

Model / Family	Considered Purpose	Documented Limitation / Observation
T5	Text generation	Would require suitable hosting and prompting/fine-tuning.
BART	Text generation	Considered for generated workout/nutrition text.
DistilBERT	Potential intent classification	Considered for classification rather than generation.
Bloom	General-purpose generation	Considered for general text generation.
Gemini	Production generation	Selected for structured and context-aware generation.

6.2 Selected AI Direction

The documentation identifies Gemini 1.5 Pro for structured 7-day workout generation and feedback-based plan updating, while Gemini Flash is identified for fast nutrition-tip generation.

6.3 Technical Architecture Idea

Layer	Technology	Main Responsibility
Frontend	HTML + Jinja2	User input and result display.
Backend	FastAPI	Routing, form processing and application logic.
AI	Gemini models	Workout, nutrition and plan-update generation.
Database	SQLite + SQLAlchemy	Persistent user and plan storage.
Server	Uvicorn	Run the FastAPI application.

7. Concept Refinement

After brainstorming individual features, the ideas were refined into one integrated concept. The key refinement was to connect the features into a continuous user journey.

7.1 From Individual Ideas to One System

The workout generator becomes the central feature. Nutrition or recovery tips support the generated plan. Feedback makes the plan adaptable. Database storage preserves the generated information, while the administrative page provides an overview of users and plans.

7.2 Refined User Flow

1. Open the FitBuddy home page.
2. Enter personal and fitness information.
3. Submit the form.
4. Generate the personalized 7-day workout plan.
5. Generate and display a nutrition or recovery tip.
6. Review the generated plan.
7. Submit feedback if changes are needed.
8. Generate an updated plan.
9. Store the original and updated information.
10. Allow administrative viewing of stored records.

7.3 Concept Definition

FitBuddy is a web-based AI fitness assistant that generates personalized 7-day workout plans and goal-related nutrition or recovery tips, while allowing users to refine their plans through feedback.

8. Ideation Output and Expected Experience

The final ideation output is a user-centered application concept with clearly connected functions. The expected experience is designed around simple input and understandable output.

8.1 Expected Home Page

The documentation describes a home page containing fields for name, user ID, age, weight, fitness goal and workout intensity. A Generate Plan action starts the AI workflow.

8.2 Expected Result Page

The result page is expected to show user information, the personalized workout plan, a nutrition tip and a feedback mechanism. The workout is organized by day and can contain exercises, sets, reps or duration, rest intervals and warm-up/cooldown guidance.

8.3 Expected Feedback Experience

The feedback page allows the user to provide their User ID and feedback about effectiveness, difficulty or preferences. The feedback is used to update the existing plan.

8.4 Expected Admin Experience

The administrative view displays user information and their original and updated workout plans. This supports monitoring of the stored application records.

8.5 Success Criteria for the Ideated Concept

- The user can provide the required fitness information.
- A structured 7-day plan can be generated.
- A related nutrition or recovery tip can be displayed.
- Feedback can be used to update the plan.
- Original and updated plans can be stored.
- Users and plans can be viewed through the administrative interface.

9. Final Brainstorming and Ideation Summary

The brainstorming and ideation phase transformed the initial problem of providing personalized fitness guidance into the FitBuddy project concept. The process identified user needs, generated possible solutions, evaluated scenarios and combined the strongest project requirements into one integrated application idea.

9.1 Final Ideas Carried Forward

- Personalized workout plan generation
- Goal and intensity-based customization
- Seven-day structured planning
- Nutrition or recovery support
- Feedback-based plan updating
- Database storage of user and plan information
- Administrative viewing of plans
- Simple web-based user interaction

9.2 Final Technology Direction

The documented technical direction uses FastAPI for backend logic and routing, HTML and Jinja2 for the frontend, Gemini models for AI-powered generation, and SQLite with SQLAlchemy for persistent storage. The project workflow can then proceed from model and architecture selection to core functionality, route development, frontend development, testing and deployment.

9.3 Conclusion

The final ideation result is a clear, modular and user-focused FitBuddy concept. Its central idea is to combine AI-generated fitness planning with supporting nutrition or recovery guidance and feedback-based adaptation. The brainstorming phase therefore provides a strong foundation for the subsequent design and implementation stages described in the project documentation.

END OF BRAINSTORMING AND IDEATION PHASE